

Stab Wound to Neck



Section I: Scenario Demographics

Scenario Title:	Stab Wound to Neck		
Date of Development:	30/09/2010 (DD/MM/YYYY)		
Target Learning Group:	☒ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Cheryl ffrench
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To review the presentation and management of neurogenic shock.
CRM Objectives:	Effectively lead a trauma team.
Medical Objectives:	1) Perform a safe and thorough primary survey. 2) Recognize the potential for neurogenic shock and initiate appropriate management. 3) Demonstrate the ability to work through the broader differential diagnosis of shock in trauma while appropriately assuming and treating hypovolemia. 4) Recognize the risk of a difficult airway in a patient with a stab wound to the neck.

Case Summary: Brief Summary of Case Progression and Major Events

A 21 year old male is brought to your tertiary care ED by EMS after being stabbed by a friend. EMS initiated spinal precautions and failed several attempts to intubate en route. On arrival, the patient is being bagged and has a single stab wound to the right posterolateral neck. He requires emergent intubation for airway protection. After intubation, his blood pressure drops but his heart rate remains in the 70s. His blood pressure will stabilize only after appropriate fluid resuscitation and vasopressor initiation.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.



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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates		Brief Description of Role	
EMS attendant		Bags the patient while giving handover.	
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☒ Blood Products	☒ ET Tubes	☒ Other: Spine board and collar	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Stab wound to the right posterolateral neck that is covered with a dressing under a C-spine collar. Blood on clothes.			
E. Approximate Timing			
Set-Up:	10 min	Scenario:	12 min
		Debriefing:	20 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

To be stated by EMS: "This is Jamal James. He's a 21 year-old male who was found in his house by police after being stabbed by a friend. There was a lot of blood at the scene. We found a stab wound on his neck so we initiated spinal precautions. Before we arrived, the police started CPR briefly because they thought he didn't have a pulse. He had a pulse when we got there but his respiratory effort was poor and he had a decreased LOC. Several attempts to intubate were unsuccessful so we bagged him on the way here. We don't know anything about his allergies, medications, or past medical history.

B. Patient Profile and History

Patient Name: Jamal James	Age: 21	Weight: 70kg
Gender: ☒ M ☐ F	Code Status: Presumed full.	
Chief Complaint: Stab wound to neck		
History of Presenting Illness: Found by police with stab to neck. Police briefly did CPR because they thought he had no pulse. Now has a pulse but decreased LOC and poor respirations.		
Past Medical History:	Unknown.	Medications: Unknown.
Allergies: Unknown.		
Social History: Unknown.		
Family History: Unknown.		
Review of Systems:	CNS:	Unknown.
	HEENT:	Unknown.
	CVS:	Unknown.
	RESP:	Unknown.
	GI:	Unknown.
	GU:	Unknown.
	MSK:	Unknown.
INT:	Unknown.	

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 80/min	BP: 110/70	RR: 6/min
O ₂ SAT: 98% on NRB		
Rhythm: Sinus	T: 36.4°C	Glucose: 5.8 mmol/L
GCS: 3 (E1 V1 M1)		
General Status: Unresponsive with agonal respirations. Being bagged by EMS, boarded and collared.		
CNS:	PERLA, 3mm bilaterally.	
HEENT:	No signs head injury. Collared with stab wound beneath collar to R posterolateral neck.	
CVS:	Normal.	
RESP:	Bags easily. Agonal respirations.	
ABDO:	Soft, NT.	
GU:	No bulbocavernous reflex.	
MSK:	No other signs injury	SKIN: Blood-stained, no other injuries.

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus HR: 80/min BP: 110/70 RR: 6/min agonal O ₂ SAT: 98% NRB T: 36.4°C	Boarded and collared. GCS 3.	<u>Learner Actions</u> - ☐ Monitors (plus defib pads) - ☐ ABCs with continued BVM - ☐ Establish IV access - ☐ Send trauma labs - ☐ Check cap sugar (5.8) - ☐ Try naloxone x1 dose - ☐ Prepare for intubation	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Naloxone given → No response - Pause bagging → O ₂ SAT to 88% <u>Triggers</u> <i>For progression to next state</i> - Intubation → 2. Intubation - 6 minutes → 4. Hypotension
2. Intubation Vitals unchanged.	As above.	<u>Learner Actions</u> - ☐ Recognize difficult airway - ☐ Call for backup, prepare adjuncts (± double set-up) - ☐ Prepare RSI meds - ☐ Apneic oxygenation - ☐ Inline stabilization - ☐ Intubate patient	<u>Modifiers</u> <u>Triggers</u> - Intubated → 3. Stabilization
3. Stabilization RR → 12 (vent) O ₂ SAT → 100%	Intubated and ventilated.	<u>Learner Actions</u> - ☐ Complete exam (D+E) - ☐ Td 0.5ml IM - ☐ CT head & CTA neck - ☐ Call ICU - ☐ CXR - ☐ Insert foley	<u>Modifiers</u> <u>Triggers</u> - 2 minutes into state → 4. Hypotension
4. Hypotension HR → 60 BP → 80/40	Intubated and ventilated.	<u>Learner Actions</u> - ☐ IV NS bolus - ☐ Call for blood - ☐ Assess for PTX - ☐ Perform FAST (normal) - ☐ Start vasopressor - ☐ ±TXA 1g iv	<u>Modifiers</u> - 2 min → BP 70/35 - 4 min → BP 60/30 - No change with fluids/blood - If not considering neurogenic shock by 4 min into state → EMS attendant re-appears and asks "What did you think of that neck wound? Woah! Do you think it's causing that low BP?" <u>Triggers</u> - Vasopressor started → 5. Resolution - 12 min → 5. Resolution
5. Resolution HR → 70 BP → 100/60	Unchanged	<u>Learner Actions</u> - ☐ Maintain vasopressor infusion - ☐ Call spine service	End Case → CT is ready!

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

No blood work is provided during the case.

Images (ECGs, CXRs, etc.)

CXR – post-intubation (normal)

Source: <https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg>

Ultrasound Video Files (if applicable)

Normal pericardial scan (no effusion)

Image credit: McMaster PoCUS Subspecialty Training Program

Normal RUQ (no free fluid)

Image credit: McMaster PoCUS Subspecialty Training Program

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) What factors in this patient's presentation were concerning for a difficult airway? 2) Did you identify neurogenic shock as a possibility? What pointed you in that direction? What is the difference between neurogenic and spinal shock? 3) What is your approach to hypotension in a trauma patient? 4) An emergent intervention (intubation) was required very early in the case. How was the team's communication surrounding this? The patient stabilized following intubation but then his blood pressure dropped. How was the team's communication surrounding this? 5) Did you feel that your team leader had control over the room? Did he or she ask for input?	
Key Moments	
Recognition and management of a difficult airway.	
Recognition and management of neurogenic shock.	
Reassessment of a trauma patient with hypotension.	

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